

The Use of Video Modeling With the Picture Exchange Communication System to Increase Independent Communicative Initiations in Preschoolers With Autism and Developmental Delays

Focus on Autism and Other
Developmental Disabilities
27(1) 3–11
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DOI: 10.1177/1088357611428426
<http://focus.sagepub.com>


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Abstract

The use of video modeling (VM) procedures in conjunction with the picture exchange communication system (PECS) to increase independent communicative initiations in preschool-age students was evaluated in this study. The four participants were 3-year-old children with limited communication skills prior to the intervention. Two of the students had been diagnosed with autism and two students exhibited developmental delays. An alternating treatments design was used to examine the effects of using VM as a priming technique to enhance the efficacy of students acquiring PECS and increasing the number of independent communicative initiations. Based on the data, the authors concluded that all students learned to use PECS and increased the number of independent communicative initiations; however, the students' rate of learning was quicker when using VM. Conclusions are discussed in the context of using empirically based interventions to teach communication skills to students with disabilities and limited verbal skills.

Keywords

video modeling, autism, communication, PECS

Developmental stages for typically developing children can be categorized by motor, sensory, cognitive, social, and communication milestones. Within the 1st year of life, infants demonstrate skills in each of these areas such as responding to the sound of a voice at 3 months of age and cooing and babbling at 6 months (Powell & Smith, 1994). Later, vocalizations develop into words with meaningful associations for people and objects. Children with a developmental delay exhibit a deficit in one of the five developmental domains of physical, cognitive, communication, social or emotional, and adaptive skills (Office of Special Education Programs, 2004). Individuals with developmental delay have highly variable characteristics as learners and communicators depending on the affected area of development; however, communication delays are common in children with developmental delays. Assistive technology, direct instruction, and modeling techniques are often used to teach communication and skill acquisition to children with developmental delays (Biederman & Freedman, 2007).

Early childhood autism is characteristically similar to developmental delay in that children do not reach typical

developmental milestones. Autism spectrum disorder (ASD) is defined as a range of complex neurodevelopmental disorders, characterized by social impairments, communication difficulties, and restricted, repetitive, and stereotyped patterns of behavior (National Institute of Neurological Disorders and Stroke, 2009). Additional characteristics include differences in sensory systems, motor skills, behavior, cognition, and rigidity of thinking (Taylor, Smiley, & Richards, 2009). Early signs of autism include lack of babbling or pointing by age 1, no single words by 16 months, and no two-word phrases by 2 years of age (Filipek et al., 2000). In addition, children with autism may exhibit no response to being addressed by name, loss of language or social skills, poor eye contact, lining up toys, no smiling or social responsiveness, and echolalia or repetition of words

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or phrases (National Institute of Neurological Disorders and Stroke, 2009). As a result of limited verbal communication, many children with ASD are unable to express emotions, make requests, or initiate communication. In many instances, this leads to frustration and inappropriate behaviors (Sterling-Turner & Jordan, 2007).

The Council of Exceptional Children's Division for Early Childhood recommends teaching communication in a natural environment by using peer models and proximity, responsive and imitative adults, and expansion of play in the resulting learning opportunities (Wolery, 2000). The National Research Council (2001) advocated teaching communication to children with autism in more naturalistic settings to improve communication, speech, and language. Communicative attempts should be initiated by the child, focused on an area of interest, embedded in an authentic setting, and followed by natural reinforcers (National Research Council, 2001). Incorporating communication interventions into the child's natural environment is the key to successful demonstration, generalizability, and maintenance of skills.

Picture Exchange Communication System (PECS)

One communication intervention that is versatile, inexpensive, and effective is the PECS (Frost & Bondy, 2002). With PECS, individuals use pictures as tools to communicate, make requests, and express ideas. The three main goals of PECS are for learners to independently initiate communication, find and approach a communicative partner, and use pictures to deliver a clear message (Preston & Carter, 2009). PECS is implemented in six phases. In Phase 1, learners exchange a single picture card for a desired object. Ultimately, in Phase 6, individuals make comments and answer questions (Frost & Bondy, 2002). PECS is an appropriate intervention for individuals with ASD in that it eliminates requirements of other communication interventions such as imitation, eye contact, and teacher prompts (Bondy & Frost, 2002).

PECS is an effective communication system for children with developmental disabilities and autism due to the inherent simplicity, lessened demands, built-in reinforcers, and versatility (Sigafoos, 2005). Tien (2008) evaluated the effectiveness of PECS by reviewing 27 published studies. Of 394 participants, only one was unable to master at least Phase 1 of PECS. In another study, Sigafoos (2005) noted that graphic-mode communication systems are more effective when teaching students with developmental delays. Ganz and Simpson (2004) also discovered that PECS increased spoken words in children with developmental delays and autism as well as increased complexity and length of utterances. PECS has been found to increase

appropriate social-communicative behaviors (Charlop-Christy, Carpenter, Le, LeBlanc, & Kellet, 2002).

Video Modeling (VM)

Another effective technique for teaching skill acquisition to individuals with autism and developmental delays is VM. VM uses a video clip that depicts a representational model engaging in a desired behavior (Bellini & Akullian, 2007). Supported by Bandura's (1977) social learning theory, VM capitalizes on the ability to learn through observation. In a VM intervention, a student watches an actor complete a target skill or task on video prior to personally engaging in the skill. Video self-modeling (VSM) adds the component of using the participant as the actor in the video.

VM is considered an effective intervention technique for children affected by autism for several reasons (Dowrick, 1999). Visual supports such as VM are appropriate for children with ASD who often demonstrate visual perception strengths (Quill, 1997). Children with ASD have an easier time attending to video-based instruction due to the visual prompts and lessened social requirements. VM also uses positive examples to demonstrate new skills (Buggey, 2007; Coyle & Cole, 2004).

VM has been used to teach communication, social, academic, and adaptive skills to individuals with autism and developmental delays (Huang & Wheeler, 2006). VM interventions have been used to teach prevocational and functional skills (Charlop, Gilmore, & Change, 2008; Cihak, Fahrenkrog, Ayres, & Smith, 2010; Delano, 2007) and social expressions (Charlop, Dennis, Carpenter, & Greenberg, 2010). Using Bandura's (1986) theories on the importance of attention on learning as a framework, Coyle and Cole (2004) used reinforcement and VM to teach on-task working behaviors and self-management to three students with ASD, resulting in decreased off-task behaviors in all participants following the intervention.

Cihak, Kessler, and Alberto (2008) used a video iPod to improve transitional behaviors for students with ASD through the use of VM, whereas Ayres, Maguire, and McClimon (2009) used video instruction to teach acquisition and generalization of chained tasks. VM promotes maintenance and generalization of skills in participants with a wide range of age and ability levels (Buggey, 2007; Delano, 2007).

In this study, we sought to evaluate the use of PECS in conjunction with VM as a priming technique to increase independent communicative interactions. Specifically, we wanted to know if the use of video priming prior to the implementation of PECS decreased acquisition time when teaching communication skills to four preschool children with autism and developmental delays. We also wanted to determine if the use of PECS in conjunction with VM was socially valid and acceptable.

Method

Participants

Four preschool children with disabilities, one special education teacher, one paraprofessional, and one teaching intern participated in this study. All children who participated in the study received direct special education services, which included 4 days a week in a developmental preschool classroom. All four children received 4 hr per week of inclusion instruction within the general education preschool setting as well. None of the children had been exposed to PECS or VM previously.

Amy. Amy was a 3-year-old girl diagnosed with autistic disorder by a developmental psychologist not associated with this study. When evaluated for academic readiness using the *Battelle Developmental Inventory–Second Edition* (BDI-2; Newborg, 2004), Amy's standard scores in the personal, social, communication, and cognitive subtests placed her in the exceptional range. On the *Vineland Adaptive Behavior Scale–Second Edition* (VABS-II; Sparrow, Cicchetti, & Balla, 2005), Amy tested in the exceptional range for the subtests of communication, socialization, and motor skills. She used no words at school, although it was reported that she had some spontaneous single-word utterances at home.

Ben. Ben was a 3-year-old boy found to be eligible for special education services due to developmental delays. Ben's standard scores in the adaptive, motor, personal-social, cognition, and communication subtests of the BDI-2 placed him in the first to fourth percentile within the exceptional range. He also received a 9- to 12-month-old age equivalency on the VABS-II. He scored within the first percentile in all subtests, including communication, daily living skills, socialization, and motor skills. Ben babbled with more than one syllable but used no words for communication.

Carl. Carl was a 3-year-old boy diagnosed with autistic disorder by a developmental psychologist not associated with this study. He received a standard score of 80 (with age equivalent of 24 months) on the *Bayley Scales of Infant and Toddler Development* (Bayley, 2006). He scored a raw score of 112 on the *Gilliam Autism Rating Scale–Second Edition* (GARS-II; Gilliam, 2006), which placed him in the "very likely" category of autism. He also received a raw score of 64 (first percentile) on the VABS-II. Carl used no words to communicate but made requests by pointing at or leading others to the desired object.

Doug. Doug was a 3-year-old boy found eligible for special education services due to a developmental delay. He received a standard score of 50 (with an age equivalent of 1 year) on the *Preschool Language Scale–Fourth Edition* (Zimmerman, Steiner, & Pond, 2002). He also scored a standard score of 4 (age equivalent of less than 30 months) on the *Early Learning Accomplishment Profile* (E-LAP; Glover, Preminger, & Sanford, 1995) of

cognitive skills. Doug did not use verbal communication but was beginning to request preferred objects through pointing.

Setting

The study occurred in an elementary school in a small public school system in the southeastern United States. More than 650 students, preschool through fourth grade, were enrolled and 50 faculty and staff members were employed at the school. The school was structured in an open pod system, with four to five classrooms in each pod. Participants in this study were enrolled in a special education developmental preschool classroom with a total of 8 children. One certified special education teacher, one paraprofessional, one nurse, and one intern were assigned to this classroom.

Data also were collected within three inclusive preschool classrooms during center time. Amy and Ben were in the same classroom, Carl was in a second classroom, and Doug was in a third classroom. Each classroom was assigned one certified teacher and one paraprofessional with class sizes ranging from 15 to 18 students.

Materials

Primary and secondary reinforcers selected via a preference assessment were identified for each child to be presented during breakfast/snack time, playtime, or center time. Preferred primary reinforcers included Fruity Pebbles, Frosted Flakes, Goldfish, Dinosaur Gummies, milk, and juice. Toy cars, dinosaurs, and Koosh balls were identified as secondary reinforcers. Boardmaker software was used to create pictures representing each of the preferred foods and toys. The picture icons were attached to a notebook with Velcro to encourage mobility and versatility. Three 20 s to 30 s videos of a peer model appropriately using PECS with a teacher were filmed using a digital flip video camera. One video clip was filmed using primary reinforcers during breakfast/snack, and the other two were filmed using secondary reinforcers in playtime and center-time areas, respectively.

Variables and Data Collection

The dependent variable was the number of independent initiations made by the student. An independent initiation was defined as the student picking up the picture and giving it to the teacher without any teacher assistance or prompt. Event recording procedures were used to record the number of independent initiations. Each student was given 10 opportunities to indicate they wanted the food/item by giving the teacher a picture. The total number of independent initiations was then divided by the total number of trials (i.e., 10) to calculate a percentage of independent initiations. Students participated in two interventions to acquire independent initiations, which included PECS only and PECS plus VM.

Table 1. Interventions and Materials for Each Student

	PECS only	PECS + VM
Food	Amy	Carl
	Ben	Doug
Item	Carl	Amy
	Doug	Ben

Note: PECS = picture exchange communication systems; VM = video modeling.

The more efficient intervention was defined as the intervention in which a student first acquired 100% independent initiation across three consecutive sessions.

Experimental Design

An alternating treatments design (Kennedy, 2005) was used to determine the efficacy of PECS in conjunction with VM. The alternating treatment design allowed us to evaluate the relationship between PECS only and PECS plus VM to assess the acquisition of independent initiations. Interventions were randomly presented to reduce potential carryover effects. Food and items were counterbalanced across students to reduce the likelihood of independent initiations occurring more often for a specified material. Table 1 displays the interventions and materials (food, item) used for each student.

Experimental Procedures

Baseline. Baseline data were collected for three sessions. During baseline conditions, students were provided 10 opportunities to exchange a picture card with a teacher to initiate the request of a desired item. The interning teacher and student sat facing one another at a half-circle table or on the floor. The relevant picture, either on the table or on a Velcro strip on a notebook, was positioned directly in front of the child within easy reach. The interning teacher placed a highly desired item (snack or toy) about 3-feet away from the child and held out her hand, palm up. A wait time of 30 s was provided to allow the student time to independently place a picture card in the interning teacher's outstretched hand. During this time, no verbal or physical prompts were provided by the teacher.

Intervention. The intervention was implemented in the special education classroom and the general education preschool classrooms. Students received either PECS instruction only or VM prior to PECS.

During the VM plus PECS intervention, the child was shown one of the three available videos of a peer independently using a picture card to request a desired item from the teacher. The peer model was a 4-year-old girl without disabilities, who was in one of the classrooms and selected based

on convenience. A laptop computer was used to play the videos for the children. The same three videos were used for all four children, but the choice of video was determined based on the setting of the upcoming activity. The child watched the video in the same natural environment as the activity. Immediately following VM procedures, PECS instruction began as described in the PECS-only intervention.

During the PECS-only intervention, the interning teacher was identified as the communicative partner or "receiver" who received the child's request. The student or "requestor" used the picture symbols to obtain desired items/foods, and the special education classroom teacher or paraprofessional served as the "assistant." The requestor and receiver sat facing one another at a half-circle table or on the floor. A single picture, either on the table or on a Velcro strip on a notebook, was positioned directly in front of the child within easy reach. The assistant sat behind the requestor, ready to provide physical prompting when necessary.

The receiver placed a highly desired item specific to the activity about 3-feet away from the child and extended one hand with the palm facing up. A wait time of 30 s was given to allow the child time to independently initiate the request with a picture. If the child displayed an inappropriate request for the item, such as grabbing, or the wait time expired, the assistant provided a physical prompt to help the child pick up the picture card and place it in the receiver's empty hand. The receiver responded by giving the item to the requestor, providing the student with positive reinforcement.

Replication. After the student reached criteria (100% independent initiations for three consecutive sessions), the preferred strategy was replicated using the materials of the less effective strategy.

Interobserver Agreement (IOA)

Data were collected by the two "assistants." The special education teacher collected data during all of the sessions, and the special education paraprofessional collected data simultaneously during 50% of the baseline sessions and a least 50% of the alternating treatment and replication phases for each student. The special education teacher and paraprofessional were trained in event recording data collection procedures and each observer followed the same sequence when recording student behavior. Percentage of IOA was calculated for each student by summing the number of agreements (either presence or absence of independent initiations) and dividing by the total number of agreements plus disagreements, and then multiplying by 100%. The percentage IOA ranged from 89% to 95% ($M = 93\%$) across all students and all phases. Amy's IOA ranged from 93% to 98% ($M = 96\%$), Ben's ranged from 87% to 100% ($M = 92\%$), Carl's ranged from 87% to 95% ($M = 89\%$), and Doug's ranged from 88% to 100% ($M = 95\%$).

Treatment Integrity

The mean treatment integrity verified the receiver's and assistant's performance and was assessed throughout all phases. The special education teacher recorded the adults' behavior for a minimum of 40% of the sessions during each phase. Intervention steps that had to be carried out by the adults consisted of (a) presenting items and snacks, (b) having pictures of snacks and items available, (c) providing access to snack and items contingent on initiations, (d) showing the corresponding video, and (e) maintaining wait time. Procedural integrity was calculated by dividing the number of observed behaviors by the number of planned behaviors and multiplying by 100%. The overall mean treatment integrity was 95% (range = 91%–100%). Amy's treatment integrity ranged from 91% to 98% ($M = 96\%$), Ben's ranged from 95% to 99% ($M = 97\%$), Carl's ranged from 90% to 100% ($M = 94\%$), and Doug's ranged from 91% to 97% ($M = 95\%$). The step that was most troublesome was when the assistant provided prompting prior to the expiration of the 30-s wait time.

Results

None of the students independently initiated picture exchange when presented snacks (food) or toys (items) during baseline. During the PECS-only intervention, all students improved independent initiations to a mean of 53.9% (range = 0%–100%) when presented the opportunity to request a snack or a toy. During PECS plus VM intervention, all students improved independent initiations to a mean of 77.6% (range = 30%–100%) when given the opportunity to request a snack or a toy. The overall mean student performance indicated that PECS plus VM was more effective. On average, students required 7.25 sessions to reach criteria. All students reached criteria during the PECS plus VM intervention, whereas none of the students reached criteria during the PECS-only intervention. Although two of the four children (Ben and Doug) were approaching criteria during the PECS-only intervention, all students acquired independent initiations quicker when VM was added to the PECS procedures. In the replication phase, PECS plus VM was used to allow students to request snacks or toys that were previously not acquired during the PECS-only condition. All of the students continued to increase independent initiations to a mean of 95.4% (range = 70%–100%) and reached criteria after 5.5 sessions.

Amy

Amy did not independently use picture cards to request snacks or toys during baseline. During the PECS-only intervention, Amy independently exchanged pictures cards a mean of 58.3% (range = 20%–80%) of the time in the

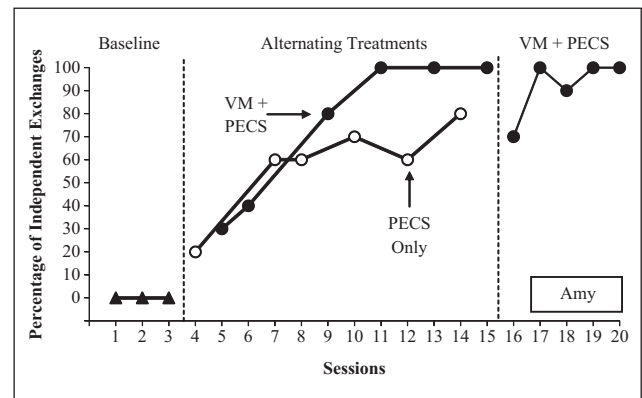


Figure 1. Amy's percentage of independent exchanges with and without VM

Note: VM = video modeling; PECS = picture exchange communication system.

presence of a snack. During the PECS-plus-VM intervention, Amy independently requested toys using picture cards 75% (range = 30%–100%) of the time. Amy reached criteria for independent initiations after six PECS-plus-VM sessions. A clear fractionization of the two intervention data paths (sessions = 9–15) occurred indicating that Amy acquired independent initiations more efficiently during the PECS-plus-VM intervention. During the replication phase, Amy reached criteria after 6 sessions for requesting food ($M = 92\%$; range = 70%–100%; see Figure 1).

Ben

Ben did not independently exchange picture cards in the presence of snacks (food) or toys (items) during baseline. During the PECS-only intervention, Ben independently initiated a picture exchange 75.7% of the time (range = 30%–100%) in the presence of food. During the PECS-plus-VM intervention, Ben independently initiated picture exchange a mean of 82.5% of the opportunities (range = 50%–100%) in the presence of a toy, reaching criteria after seven PECS-plus-VM sessions. Ben performed slightly better during the PECS-plus-VM intervention, although outcomes for both interventions were functionally similar. During the replication phase, Ben increased independent initiations for food with a mean of 97.1% (range = 90%–100%) and reached criteria after seven sessions (see Figure 2).

Carl

Carl did not independently initiate picture exchanges in the presence of either snacks (food) or toys (items) during baseline. During the PECS-only intervention, Carl independently initiated picture exchange to a mean of 32% (range =

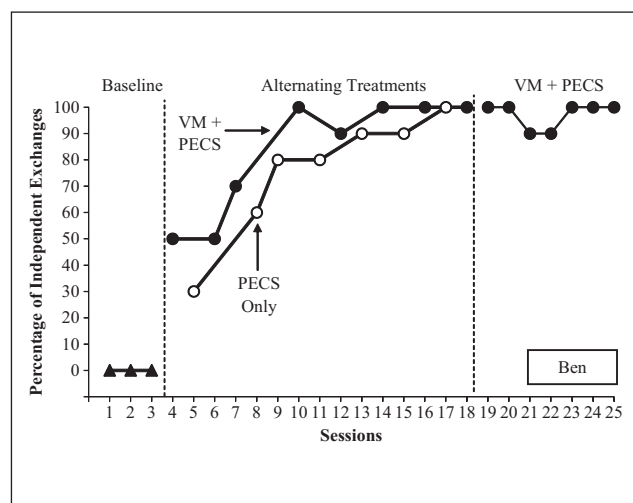


Figure 2. Ben's percentage of independent exchanges with and without VM

Note: VM = video modeling; PECS = picture exchange communication system.

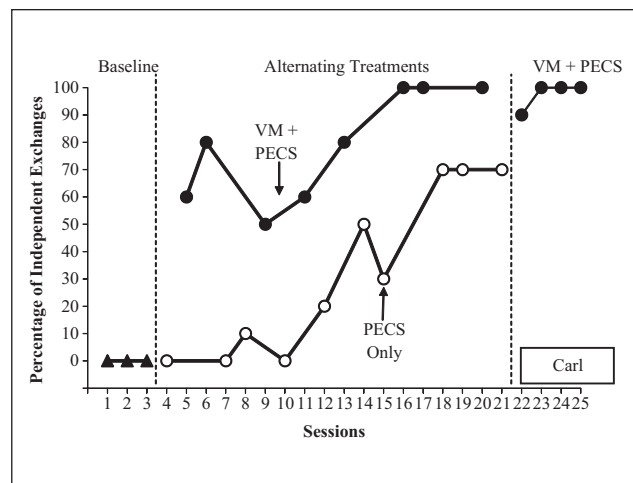


Figure 3. Carl's percentage of independent exchanges with and without VM

Note: VM = video modeling; PECS = picture exchange communication system.

0%–70%) in the presence of a toy. During the PECS-plus-VM intervention, Carl independently initiated picture exchange 78.8% of the opportunities to gain a snack (range = 50%–100%) and reached criteria after eight PECS-plus-VM sessions. A clear fractionization of the two intervention data paths occurred throughout initial acquisition indicating that Carl acquired independent initiations more efficiently during the PECS-plus-VM intervention. During the replication phase, Carl independently initiated requests for toys with a mean of 97.5% (range = 90%–100%) and reached criteria after four sessions (see Figure 3).

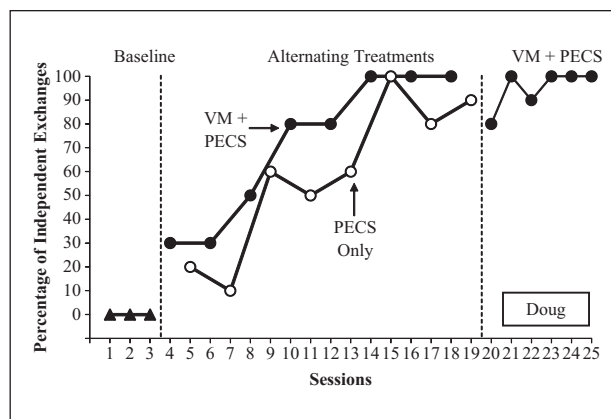


Figure 4. Doug's percentage of independent exchanges with and without VM

Note: VM = video modeling; PECS = picture exchange communication systems.

Doug

Doug did not independently initiate picture exchange requests for snacks (food) or toys (items) during baseline. During the PECS-only intervention, Doug improved independent initiations to a mean of 58.8% (range = 0%–100%) in the presence of a toy. During PECS-plus-VM intervention, Doug improved independent initiations to a mean of 71.3% (range = 30%–100%) in the presence of a snack and reached criteria after eight sessions. Doug performed slightly better during the PECS-plus-VM intervention, although outcomes for both interventions were functionally similar. During the replication phase, Doug independently initiated picture exchanges for toys with a mean of 95% of the opportunities (range = 80%–100%), reaching criteria after six sessions (Figure 4).

Social Validity

At the conclusion of the study, the special education teacher, the teacher intern, the three general education teachers, and the three general education paraprofessionals were asked to complete a social validity questionnaire. The survey was composed of four short-answer questions. Staff members were asked to fill out separate questionnaires for each child.

Results from the survey can be interpreted to conclude that all instructors believed the effects of the use of VM before PECS instruction were positive for all children. The students' teachers and paraprofessionals indicated that the opportunity to view appropriate communication in collaboration with adult support contributed to the positive effects of VM prior to PECS. As the children in this study were unable to communicate, no student reporting was included. Based on the students' data, the general and special education

teachers expressed plans to continue to use VM as an intervention method for a wide variety of skills as well as to recommend it to other teachers.

Discussion

Our purpose in conducting this study was to evaluate the use of PECS in conjunction with VM as a priming technique to increase independent communicative interactions in preschool students with limited to no verbal communication skills. Prior to the study, all of the students demonstrated limited communication abilities with no independent picture exchanges during baseline to request food or toy items. As a result of using VM as a priming technique in conjunction with PECS, all students increased independent initiations and reached criteria more quickly. The intervention was effective for teaching requests of primary (food) and secondary (toys) reinforcers. In addition, all students demonstrated the ability to communicate with others through the use of PECS, which is a functional, practical, and economical mode of communication. Ultimately, all students achieved the goal of increasing independent picture exchange initiations.

Our findings supports previous findings that suggest VM is an effective technique for teaching new skills (Buggey, 2007; Cihak et al., 2010; Dowrick, 1999) and improving communication skills (Charlop et al., 2010). In addition, our findings can be used to support other VM research in which participants demonstrated success and skill maintenance over time (Buggey, 2005). The findings of this study also support the effectiveness of PECS as a communicative tool for students with autism and developmental disabilities (Charlop-Christy et al., 2002; Ganz & Simpson, 2004; Sigafos, 2005).

The finding from our study can be used to extend the VM and PECS literature in various ways. First, VM was used as a priming technique to teach the acquisition of PECS. Previous researchers have focused on the benefits of using VM and PECS as two separate interventions to teach communication skills (Bellini & Akullian, 2007; Bondy & Frost, 2002). In this study, we combined the two methods with the emphasis on VM prior to engaging in a communicative interaction with PECS. This innovative technique resulted in quicker acquisition and mastery of criteria for all students. By viewing the video prior to engaging in a communicative interaction, students with autism and developmental delays were able to more accurately replicate the steps and actions depicted in the video to independently communicate access to desired commodities.

Second, students were able to master the communication system of PECS more quickly with VM than through the PECS-only intervention. Every person with communication deficits has the right to an alternative form of communication. PECS is a simple, effective communication

system that enables individuals to express ideas through a picture exchange. In addition to serving as a primary means of communication, PECS also has been shown to increase verbal communication, so its effectiveness is twofold (Ganz & Simpson, 2004.). Although PECS has been shown to be effective, learning any new communication system can be a lengthy and time-consuming process. By adding VM as a priming technique, students learned PECS quicker. Through this study, the acquisition time of learning PECS was demonstrated to be shorter than without the use of video priming. This will have an affect on professionals and paraprofessionals who teach this and other communication methods to students who lack the ability to communicate independently. Training time for students can potentially be reduced resulting in lower demands on teachers and paraprofessionals. A video model of a peer is easy to produce and access.

Third, we included a social validity component to the study to assess the usefulness and potential applicability of the PECS-plus-VM intervention. All three instructors indicated that this was a necessary and warranted intervention based on the communicative skill levels of the preschool students. In addition, each educator indicated that VM is an effective method of teaching PECS, that the intervention increased independent communicative interactions, and that they would be willing to teach PECS via VM in their classroom and recommend it to other teachers. The classroom teachers noted that the intervention was simple to add to a daily schedule and easy to implement in a typical school day. The educators agreed that this is a socially valid intervention technique.

Limitations

To fully interpret and apply the results of this study, several limitations should be considered. As with all single-case designs, a small sample was used; we had only four participants. Conclusions from the study should be interpreted and applied in the context of the small number of participating students. To increase external validity and generalizability of results, future researchers should include a larger sample size. It also is important to evaluate specific characteristics of the four participating students. All four students were preschoolers diagnosed with autism or found eligible for special education services based on developmental delays. The students all attended the same school and were from similar cultural and socioeconomic backgrounds. Due to these shared characteristics, results cannot be generalized to all preschoolers with disabilities or to other age groups.

Time restrictions may be considered a third limiting factor of this study. As with most school-based interventions,

this study was conducted under the time constraints of a typical school day and calendar. Extending the time to implement and evaluate the effectiveness of the intervention would allow future researchers to fully investigate the effects of the intervention. A limitation in using the same model for all students is that according to Bandura's social learning theory, using a more similar model may have offered students an opportunity to achieve mastery more quickly. Finally, it is important to note that the students participating in this study were highly motivated by salient reinforcers obtained through a preference assessment. Evaluations across a broad scope of reinforcers should be completed in the future.

Future Research

As the results of this study can be interpreted to suggest, VM may be an effective technique to use in conjunction with PECS for students with autism or developmental disabilities who have communication needs. It is necessary to evaluate this intervention with other groups of participants in the future including those of different ages and cultural backgrounds. In addition, researchers should consider using VM to teach topics from other areas of need such as academic, life, and social skills. Furthermore, researches in future studies should investigate the use of PECS and VM in community settings using natural supports. This would be especially appropriate for older students who desire increased independence and a universal form of communication.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

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